

Attendance Module

Architectural Implementation Plan — v2 Refined

Project: Printing Shop Management System (PHP Payroll)

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Technology: Laravel 12 · PHP 8.4 · SQLite/MySQL · React/Inertia · Tailwind CSS v4

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1. Architectural Strategy — 3-Layer Data Approach

We decouple raw time logs from final payroll calculations using a strict 3-layer data model. Each layer is independently verifiable, hash-verified for tamper detection, and locked once consumed downstream.

Layer	Table	Role	Mutability
1. Raw Ledger	<code>time_logs</code>	Immutable punch records: IN/OUT with timestamp, source, and optional duplicate-of self-reference. Corrections append new rows — never update existing ones.	Immutable
2. Daily Worksheets	<code>attendance_sheets</code>	Computed daily evaluation: reads <code>time_logs</code> , compares against the employee's scheduled shift, applies grace periods, overtime rules, holiday overrides, and leave blending. Each row has a <code>computation_hash</code> (SHA-256 of all inputs).	Re-computable until <code>locked_at</code> is set
3. Period Summaries	<code>payroll_period_items</code>	Aggregate summaries for a 15-day or 30-day payroll cutoff period. Generated from locked attendance sheets. Feeds the payroll run engine for final net pay computation.	Locked once period is approved/paid

Design Principle: Raw data is never overwritten. Corrections create new audit-trailed records. Computed sheets can be regenerated by re-running `processDailyAttendance()` as long as they haven't been consumed by a payroll period. Once a payroll period is approved, all constituent sheets are permanently locked — requiring full period voidal to make changes.

1.1 Supported Punch Sources

Source	Initiator	Requires Approval?	Description
self_service	Employee	No	Employee punches via their self-service portal
biometric	Hardware device	No	Automated biometric/fingerprint scanner
manual	Admin (trusted)	No	Admin directly creates a time_log (e.g., biometric was down, supervisor verified attendance in person)
correction	System (auto)	Pre-approved	Created automatically when an attendance_correction_request is approved. Links back to the request.

1.2 Duplicate Punch Throttling

Multiple punches of the same type (IN/IN or OUT/OUT) within a **5-minute window** are throttled: the **earliest** punch is kept as the canonical record. Subsequent duplicate punches are created as database rows but marked with a `duplicate_of` self-reference pointing to the canonical record. This preserves the full audit trail while ensuring computation uses only one punch per type.

2. Database Schema Design

All new tables use **UUIDs** as primary keys (`Str::orderedUuid()`) for index-friendly ordering. Foreign key references to existing tables (`employees` , `branches` , `users`) remain standard big integers to match the existing schema.

2.1 company_configurations

Key-value table for all admin-configurable system parameters. Supports branch-level overrides via optional `branch_id` FK (null = system-wide default). Typed values prevent string coercion errors.

Column	Type	Description
id (PK)	uuid	Ordered UUID, primary key
branch_id (FK)	bigint nullable	Branch override scope; null = global default
key	string(100)	Unique parameter key (unique per branch_id scope)
value	text	Parameter value as a string
type	string(20)	Coercion type: string, integer, boolean, decimal
group	string(50)	Grouping: attendance, payroll, general

Default Seed Values (Attendance Group):

Key	Default	Type	Description
grace_period_minutes	15	integer	Free late allowance in minutes
forfeit_grace_period_on_exceed	true	boolean	PH standard: forfeit full grace if exceeded
ot_threshold_minutes	60	integer	Min consecutive minutes before OT qualifies
auto_lunch_deduction_minutes	60	integer	Unpaid lunch break auto-deduction
half_day_threshold_minutes	240	integer	4 hours = half-day cutoff
late_deduction_per_minute	10.00	decimal	Pesos per late minute
late_deduction_max_minutes	30	integer	Cap on late minutes chargeable per day
ot_rate_regular	1.25	decimal	Regular day OT multiplier
ot_rate_rest_day	1.30	decimal	Rest day OT multiplier
ot_rate_holiday_regular	2.60	decimal	Regular holiday OT multiplier
ot_rate_holiday_special	1.30	decimal	Special non-working day OT multiplier

2.2 employee_schedules

Column	Type	Description
id (PK)	uuid	Ordered UUID
employee_id (FK)	bigint	→ employees.id, cascade on delete
branch_id (FK)	bigint	→ branches.id, denormalized for scoping
schedule_name	string(100)	e.g. 'Standard 8-5', 'Morning Shift'
scheduled_start	time	e.g. 08:00
scheduled_end	time	e.g. 17:00
work_days	json	[1,2,3,4,5] where 1=Monday, 7=Sunday
effective_from	date	Schedule effective start date
effective_to	date nullable	null = current/active schedule
is_active	boolean	Soft flag; can deactivate without deleting
is_default	boolean	Default schedule for new employees
rest_day_work_multiplier	decimal nullable	Override rest-day rate if called in

2.3 time_logs (Immutable Punch Ledger)

Column	Type	Description
id (PK)	uuid	Ordered UUID
employee_id (FK)	bigint	→ employees.id
branch_id (FK)	bigint	→ branches.id (denormalized)
timestamp	datetime	Precise to seconds, when punch occurred
type	string(10)	IN or OUT
source	string(30)	self_service biometric manual correction
metadata	json nullable	Device info, geo-location, admin note
duplicate_of	uuid nullable	Self-ref FK → time_logs.id for throttled dupes
created_by	bigint nullable	→ users.id
created_at	timestamp	Immutable — no updated_at column

Constraints: No `updated_at` column. No delete operations. Corrections append new rows. Indexed on `(employee_id, timestamp)`, `(branch_id, timestamp)`, and `(employee_id, DATE(timestamp), type)`.

2.4 attendance_correction_requests

Column	Type	Description
id (PK)	uuid	Ordered UUID
employee_id (FK)	bigint	→ employees.id (whose record)
requested_by (FK)	bigint	→ users.id (who submitted)
branch_id (FK)	bigint	→ branches.id
attendance_date	date	Date being corrected
correction_type	string(25)	missed_punch_in missed_punch_out time_adjustment absent_to_present
requested_time_in	time nullable	Proposed corrected IN time
requested_time_out	time nullable	Proposed corrected OUT time
original_time_in	time nullable	Old/wrong time (for time_adjustment)
original_time_out	time nullable	Old/wrong time (for time_adjustment)
reason	text	Required justification
status	string(20)	pending approved denied
reviewed_by	bigint nullable	→ users.id (approver)
denial_reason	text nullable	Required if denied
resolved_time_log_id	uuid nullable	→ time_logs.id created on approval

Approval Rules: Only one pending request per (employee_id, attendance_date, correction_type) . Submitter cannot self-approve. Submitter's role must be lower than reviewer's role (staff→admin, admin→superadmin).

2.5 overtime_requests

Column	Type	Description
id (PK)	uuid	Ordered UUID
employee_id (FK)	bigint	→ employees.id
overtime_date	date	Date of planned OT
scheduled_end	time	Snapshot of shift end for audit
requested_hours	decimal(4,2)	e.g. 2.50
requested_minutes	integer	Computed: requested_hours × 60
shift_type	string(20)	regular_day rest_day regular_holiday special_holiday
ot_rate	decimal(4,2) nullable	Snapshot of resolved rate at approval time
reason	text nullable	Business justification
status	string(20)	pending approved denied
approved_by	bigint nullable	→ users.id

2.6 leave_requests

Column	Type	Description
id (PK)	uuid	Ordered UUID
employee_id (FK)	bigint	→ employees.id
leave_date	date	Single day
leave_type	string(30)	vacation sick emergency maternity paternity bereavement unpaid
duration	string(20)	full_day half_day_morning half_day_afternoon
is_paid	boolean	Default true; false for unpaid leave
status	string(20)	pending approved denied
reason	text nullable	Employee's reason
approved_by	bigint nullable	→ users.id

2.7 holidays

Column	Type	Description
id (PK)	uuid	Ordered UUID
name	string(200)	e.g. 'Araw ng Kagitingan'
holiday_date	date	Calendar date
type	string(20)	regular special_non_working
is_recurring_yearly	boolean	Auto-roll to next year

2.8 attendance_sheets (Daily Worksheets)

Column	Type	Description
id (PK)	uuid	Ordered UUID
employee_id (FK)	bigint	→ employees.id
branch_id (FK)	bigint	→ branches.id
date	date	Unique per (employee_id, date)
schedule_id (FK)	uuid nullable	→ employee_schedules.id active at time
schedule_start	time nullable	Shift start
schedule_end	time nullable	Shift end
first_time_in	datetime nullable	Earliest IN punch
last_time_out	datetime nullable	Latest OUT punch
raw_worked_minutes	integer	Total elapsed: IN→OUT
lunch_deducted_minutes	integer	60 or 0 per auto-lunch rule
actual_worked_minutes	integer	raw - lunch
late_minutes	integer	After grace period calc
late_deduction_amount	decimal(10,2)	late_minutes × rate, capped at 30min
undertime_minutes	integer	Early departure deduction
raw_overtime_minutes	integer	Actual stay past schedule_end
approved_overtime_minutes	integer	After lower-of-two cross-reference
ot_rate_applied	decimal(4,2) nullable	Multiplier used (1.25/1.30/2.60)
ot_amount	decimal(10,2)	Computed OT earnings
regular_hours	decimal(4,2)	e.g. 8.00 or 4.00 (half-day)
is_present	boolean	Worked or on approved paid leave
is_half_day	boolean	Exactly 4 hours credit
is_absent	boolean	No pay for the day
is_rest_day	boolean	Not in schedule's work_days
is_rest_day_worked	boolean	is_rest_day AND clocked in
absence_type	string(20) nullable	unexcused approved_leave

holiday_type	string(20) nullable	regular special_non_working
holiday_pay_percent	integer	0 100 130 200 260
leave_request_id	uuid nullable	→ leave_requests.id
approved_ot_request_id	uuid nullable	→ overtime_requests.id
daily_rate	decimal(10,2)	Snapshot of employee's rate on date
daily_wage	decimal(10,2)	Computed gross for the day
computation_hash	string(64)	SHA-256 of all inputs for change detection
computed_by	bigint nullable	→ users.id
locked_at	timestamp nullable	Set when consumed by payroll period

2.9 payroll_periods

Column	Type	Description
id (PK)	uuid	Ordered UUID
branch_id (FK)	bigint	→ branches.id
period_start	date	Cutoff start
period_end	date	Cutoff end
period_type	string(10)	first_half (1–15) second_half (16–EOM) full
status	string(20)	draft computed approved paid
created_by	bigint	→ users.id
approved_by	bigint nullable	→ users.id

2.10 payroll_period_items

Column	Type	Description
id (PK)	uuid	Ordered UUID
payroll_period_id (FK)	uuid	→ payroll_periods.id
employee_id (FK)	bigint	→ employees.id
branch_id (FK)	bigint	→ branches.id
total_regular_days	decimal(4,2)	Sum of regular_hours / 8
total_half_days	decimal(4,2)	0.5 per half-day
total_absent_days	decimal(4,2)	Unexcused absences
total_late_minutes	integer	Period aggregate
total_undertime_minutes	integer	Period aggregate
total_overtime_minutes	integer	Approved OT only
total_holiday_pay_days	decimal(4,2)	Holiday day equivalents
total_leave_paid_days	decimal(4,2)	Paid leave day equivalents
daily_rate_snapshot	decimal(10,2)	Employee rate at period time
gross_pay	decimal(10,2)	Sum of all daily_wages
late_deduction	decimal(10,2)	Sum of late penalties
undertime_deduction	decimal(10,2)	Sum of undertime deductions
overtime_pay	decimal(10,2)	Sum of OT earnings
holiday_pay	decimal(10,2)	Holiday premium
government_deductions	decimal(10,2)	SSS + PhilHealth + Pag-IBIG
cash_advance_deduction	decimal(10,2)	Deducted CA
net_pay	decimal(10,2)	Final net pay

3. Enum Definitions

All string-based status/type columns use PHP Backed Enums for type safety. Enums live in `payroll/Attendance/Enums/`.

Enum Class	Values	Stored As
TimeLogType	IN, OUT	time_logs.type
TimeLogSource	self_service, biometric, manual, correction	time_logs.source
CorrectionType	missed_punch_in, missed_punch_out, time_adjustment, absent_to_present	attendance_correction_requests.correction_type
OvertimeRequestStatus	pending, approved, denied	overtime_requests.status
LeaveType	vacation, sick, emergency, maternity, paternity, bereavement, unpaid	leave_requests.leave_type
LeaveDuration	full_day, half_day_morning, half_day_afternoon	leave_requests.duration
HolidayType	regular, special_non_working	holidays.type
AbsenceType	unexcused, approved_leave	attendance_sheets.absence_type
PayrollPeriodStatus	draft, computed, approved, paid	payroll_periods.status
OTShiftType	regular_day, rest_day, regular_holiday, special_holiday	overtime_requests.shift_type

4. Business Rules & Computation Logic

4.1 Shift & Core Tracking Rules

- **No night/graveyard shifts:** All standard shifts fall cleanly within a single calendar day.
- **2-Punch Culture with Auto-Deduction:** Employees clock IN once (arrival) and OUT once (departure). The system uses only the earliest IN and latest OUT — no summing of intermediate punch intervals. This is the standard Philippine practice for daily-paid workers.
- **Only IN but no OUT:** Treated as "incomplete." If the current time is still within business hours, no sheet is generated (marked as in-progress). If the day is fully closed (past midnight or processed the next day), flagged as absent.
- **Only OUT but no IN:** Flagged as an anomaly. Cannot determine worked duration. Warning added to sheet notes.

4.2 Lunch Break Rule

The system automatically subtracts exactly **60 minutes** for the unpaid lunch break, provided two conditions are met:

1. The total raw duration (IN→OUT) is **≥ 5 hours (300 minutes)**.
2. The work period actually **spans the noon break window** (12:00–13:00).

If either condition is not met, lunch deduction is 0.

4.3 Grace Period & Lateness

Scenario	late_minutes
On time or early: $\text{time_in} \leq \text{schedule_start}$	0
Late but within grace: $\text{total_late} \leq \text{grace_period_minutes}$	0
Exceeds grace + $\text{forfeit_grace_period_on_exceed} = \text{true}$ (PH standard)	Full total_late_minutes (grace forfeited)
Exceeds grace + $\text{forfeit_grace_period_on_exceed} = \text{false}$	$\text{total_late_minutes} - \text{grace_period_minutes}$ (only excess)

Late Deduction Formula:

$\text{chargeable} = \min(\text{late_minutes}, \text{late_deduction_max_minutes})$ — capped at 30 minutes per day

$\text{deduction} = \text{chargeable} \times \text{late_deduction_per_minute}$ — e.g. $30 \times \text{₱}10 = \text{max ₱}300/\text{day}$

Early Bird Rule: Clocking in before the shift start time yields 0 regular hours or OT hours for that early period, unless an approved early OT record exists. The employee is simply marked as on time.

4.4 Undertime, Half-Day & The 3-Hour Block Penalty

Actual Worked Minutes	Classification	Pay
$\geq$ scheduled shift minutes	Full Day	100% daily rate
> 240 and $<$ scheduled shift	Partial Day with Undertime	Proportional: $\text{actual_minutes} / \text{scheduled_minutes} \times \text{daily_rate}$
$= 240$ (exactly 4 hours)	Half-Day	50% daily rate
1–239 (less than 4 hours)	Unexcused Absence	0% — 3-Hour Block Penalty
0 (no punches)	Absent	0%

The 3-Hour Rule (key in Philippine payroll): If an employee works less than 4 hours, they fail to complete a half-day block. The system automatically flags the day as an unexcused absence with 0 hours paid, **unless** an approved leave or undertime slip overrides it. Partially worked rest days are **exempt** from the 3-hour block — rest day work is paid for actual minutes at the rest day rate with no minimum threshold.

4.5 Overtime Rules

- Approval-Driven:** Actual extra time spent past the shift end time does **not** trigger auto-payment. The system cross-references actual stay duration against a supervisor-approved `overtime_request` record and awards the **lower value** of the two. Unapproved minutes are discarded.
- OT Threshold:** Admin can set a minimum threshold (e.g., 60 consecutive minutes) before extra time qualifies as valid OT. Extra time below this threshold is discarded even if an approved request exists.
- Rate Selection:** The OT rate is determined by the `shift_type` on the approved overtime request at processing time.

4.6 OT Rate Selection Matrix (PH DOLE Standard)

Day Type	Worked?	OT Multiplier	Source
Regular work day	Yes	1.25	ot_rate_regular
Rest day	Yes	1.30	ot_rate_rest_day
Regular holiday	Yes	2.60	ot_rate_holiday_regular
Special non-working day	Yes	1.30	ot_rate_holiday_special
Rest day + regular holiday	Yes	2.60	Regular holiday rate applies
Rest day + special holiday	Yes	1.50	Standard special+rest day rate

4.7 Holiday Pay Rules (PH DOLE)

Regular Holidays (Unworked)

Day-Before Rule: Entitlement to 100% daily pay requires an automated check of the **last scheduled working day** immediately preceding the holiday. The system walks backward through the calendar, skipping rest days and Sundays, to find the last working day.

Day-Before Status	Unworked Holiday Pay
Present OR Approved Paid Leave	100% daily rate
Absent without leave	0%

Regular Holiday (Worked) — 200%

Day Before	Worked Holiday Pay
Present or Leave	200% of daily rate for first 8 hours
Absent without leave	200% of daily rate for first 8 hours

Special Non-Working Days: Operates strictly on "**No work, no pay**" — 0% if unworked, 130% of daily rate if worked.

4.8 Leave Integration & Blending

Partial leaves (e.g., 4-hour morning or afternoon) blend with actual hours worked to achieve a full day's base wage:

Leave Duration	Actual Worked	Result
Full day (paid)	Any (including 0)	Full day credited, 100% paid. Absence type = approved_leave.
Half-day morning	Afternoon ≥ 4 hours	Full day blended (4h leave + 4h+ worked). No half-day flag.
Half-day morning	Afternoon < 4 hours	Insufficient. Half-day leave credited only. Worked portion insufficient for half-day block $\rightarrow$ remainder unpaid.
Half-day afternoon	Morning ≥ 4 hours	Full day blended (4h+ worked + 4h leave). No half-day flag.
Half-day afternoon	Morning < 4 hours	Insufficient. Same handling as above — 3-hour rule applies to worked portion.
Full day (unpaid)	Any	Hours credited but daily_wage = 0 for those hours.

4.9 Daily Wage Computation Formula

Regular Pay:

`hourly_rate = daily_rate / scheduled_work_hours`

`regular_pay = hourly_rate  $\times$  regular_hours`

Holiday Adjustment:

If `holiday_pay_percent > 100` (worked holiday): `regular_pay = regular_pay  $\times$  (holiday_pay_percent / 100)`

If `holiday_pay_percent = 100` AND unworked: `regular_pay = daily_rate` (100% holiday pay)

Overtime Pay:

`ot_pay = hourly_rate  $\times$  ot_rate_applied  $\times$  (approved_overtime_minutes / 60)`

Deductions:

`late_deduction = min(late_minutes, 30)  $\times$  10`

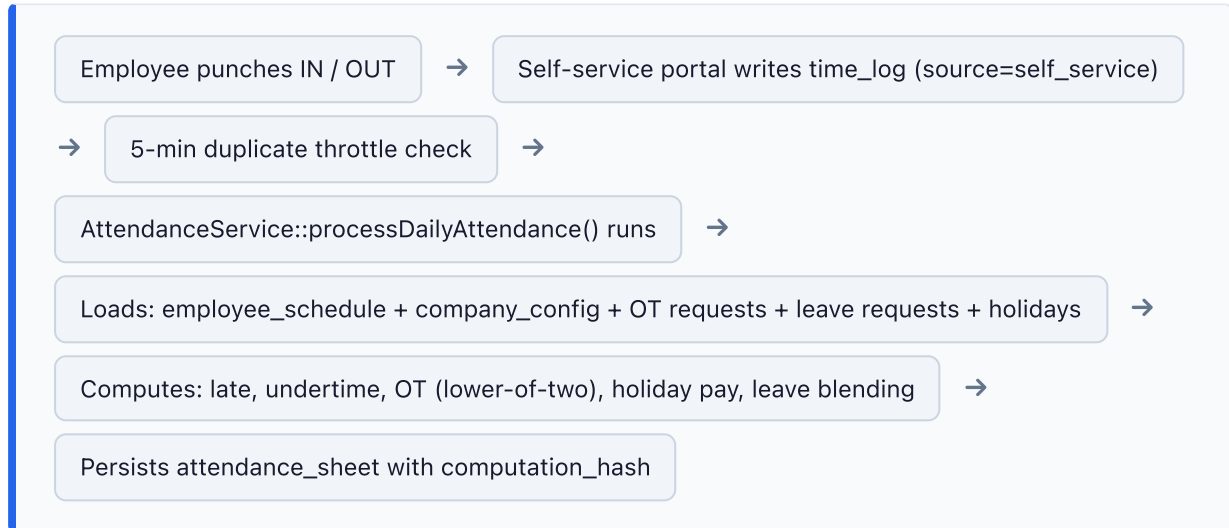
`undertime_deduction = hourly_rate  $\times$  (undertime_minutes / 60)`

Total Daily Wage:

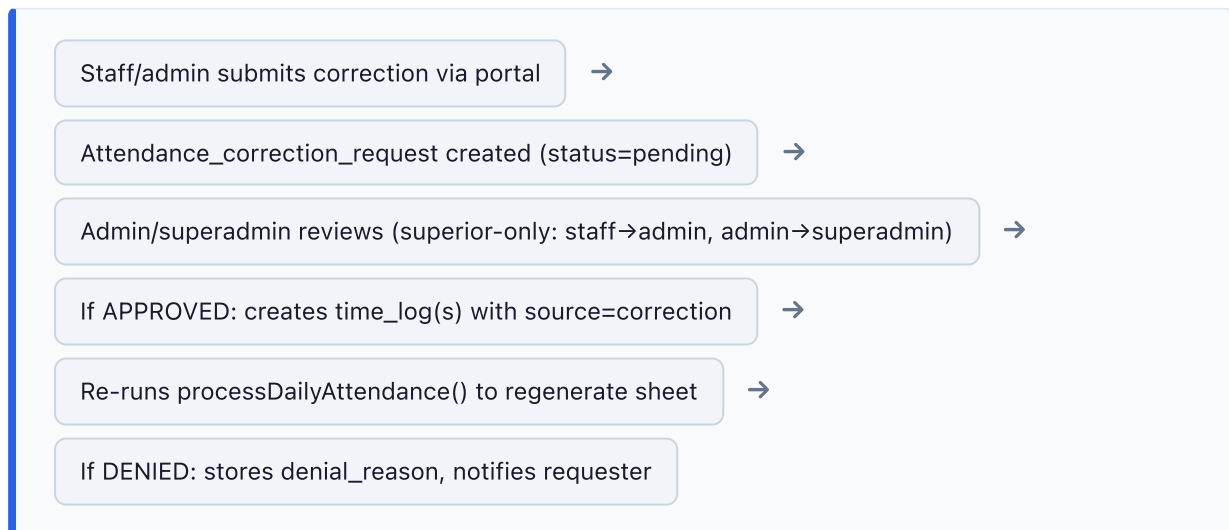
`daily_wage = regular_pay + ot_pay - late_deduction - undertime_deduction`

5. Workflows

5.1 Daily Attendance Flow

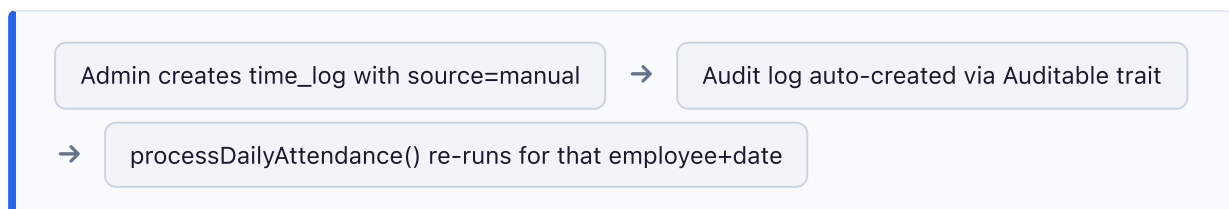


5.2 Manual Attendance Correction Flow (Approval-Gated)



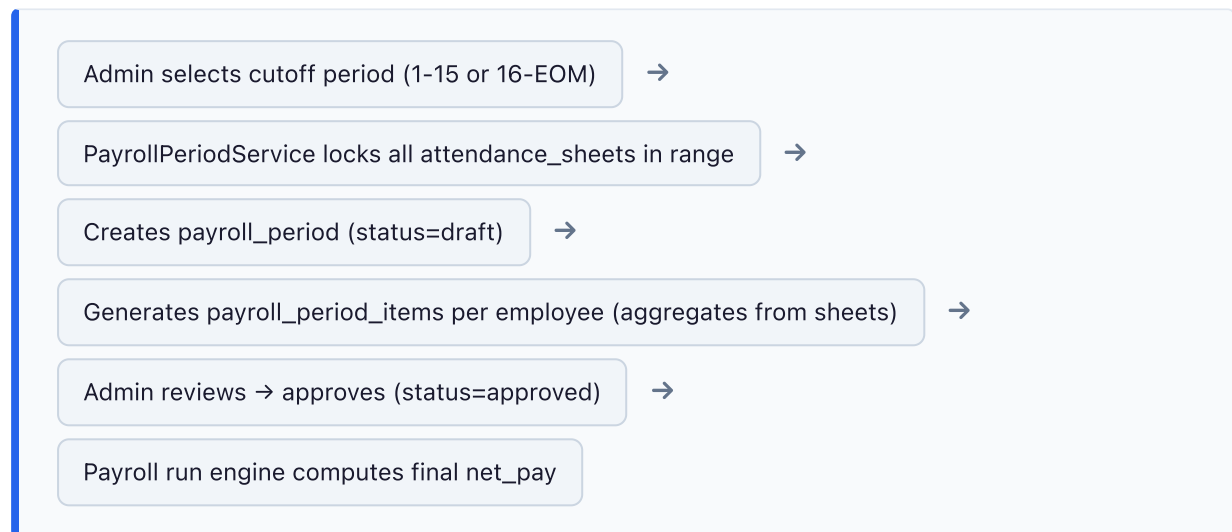
Duplicate Prevention: Only one pending correction request per (employee_id, attendance_date, correction_type) is allowed at any time.

5.3 Admin Direct Manual Log Flow (No Approval)



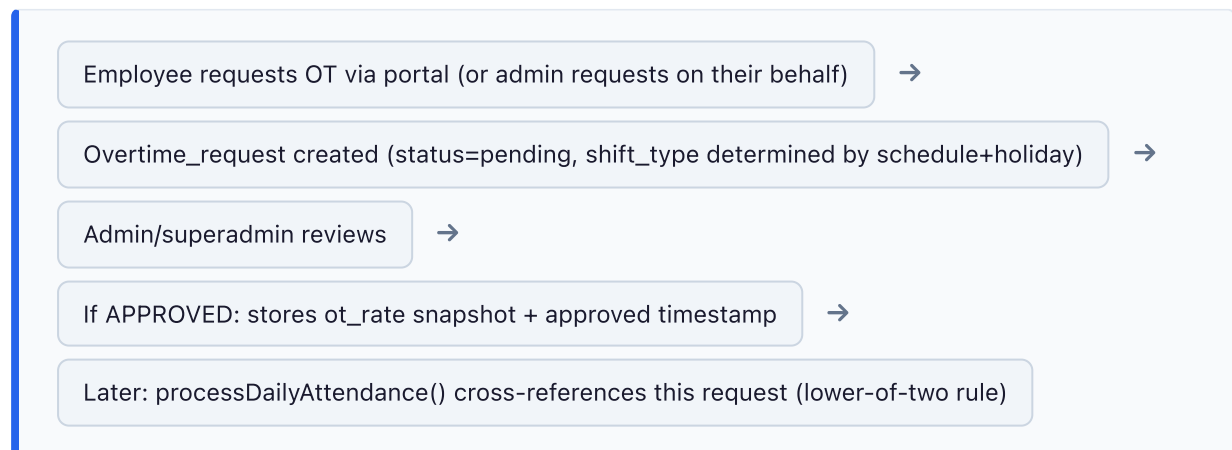
Admin manual logs are trusted (admin has authority). Only available to admin and superadmin roles. Staff cannot create manual time_logs.

5.4 Payroll Period Generation Flow



Lockdown Rule: Once attendance_sheets have `locked_at IS NOT NULL`, they cannot be recomputed or edited. To make changes, the superadmin must **void the entire payroll period** (which unlocks all constituent sheets), make corrections, then regenerate.

5.5 Overtime Approval Flow



6. Authorization Matrix

Operation	Staff	Admin (branch)	Superadmin
View own attendance / punch history	Yes	Yes	Yes
Self-service punch IN / OUT	Yes (own only)	Yes (own only)	Yes (own only)
View branch attendance sheets	No	Yes (own branch)	Yes (all branches)
Submit correction request	Self only	Self + any in branch	Self + any
Approve correction request	No	Staff submissions only (not own, not admin)	Admin + staff submissions (not own superadmin)
Create manual time_log (source>manual)	No	Yes (own branch)	Yes (all branches)
Manage employee schedules	No	Yes (own branch)	Yes (all branches)
Manage holidays	No	No	Yes
Approve OT requests	No	Yes (own branch)	Yes (all branches)
Approve leave requests	No	Yes (own branch)	Yes (all branches)
Edit company configuration	No	No	Yes
Generate payroll period	No	Yes (own branch)	Yes (all branches)
Approve payroll period	No	Yes (own branch)	Yes (all branches)
Void payroll period	No	No	Yes

All authorization is enforced through Laravel Policies; no inline role-string comparisons.

7. Route Structure

7.1 Self-Service Portal (all authenticated roles)

Method	URI	Name	Description
GET	/payroll/my-attendance	payroll.attendance.my	Employee's own attendance + punch buttons
POST	/payroll/punch	payroll.attendance.punch	Self-service punch IN/OUT

7.2 Attendance Sheets (admin/superadmin only)

Method	URI	Name	Description
GET	/payroll/attendance	payroll.attendance.index	Branch attendance sheet view
POST	/payroll/attendance/process/{branch}	payroll.attendance.process	Batch process all employees for a date

7.3 Corrections

Method	URI	Name	Description
GET	/payroll/corrections	payroll.corrections.index	Review queue (badge: pending count)
POST	/payroll/corrections	payroll.corrections.store	Submit correction (staff/admin)
POST	/payroll/corrections/{id}/approve	payroll.corrections.approve	Approve (superior only)
POST	/payroll/corrections/{id}/deny	payroll.corrections.deny	Deny with reason

7.4 Schedules, OT, Leave, Holidays (admin/superadmin only)

Resource	Prefix	Operations
Schedules	/payroll/schedules	index, store, update, destroy
Overtime	/payroll/overtime	index, store, update, destroy, approve, deny
Leaves	/payroll/leaves	index, store, update, destroy, approve, deny
Holidays	/payroll/holidays	index, store, update, destroy

7.5 Payroll Periods (admin/superadmin)

Method	URI	Name	Description
GET	/payroll/periods	payroll.periods.index	List all periods
POST	/payroll/periods/generate	payroll.periods.generate	Generate new period from locked sheets
GET	/payroll/periods/{id}	payroll.periods.show	Period detail with line items
POST	/payroll/periods/{id}/approve	payroll.periods.approve	Approve period (locks items)
POST	/payroll/periods/{id}/void	payroll.periods.void	Superadmin: void period, unlock sheets

8. Frontend Page Plan

8.1 Sidebar Navigation Update

Add **Attendance** dropdown between Employees and Audit Logs in the payroll sidebar.
Expanded items visible based on role:

Nav Item	Staff	Admin	Superadmin
My Attendance	Yes	Yes	Yes
Attendance Sheets	—	Yes	Yes
Corrections pending count	—	Yes	Yes
Schedules	—	Yes	Yes
Overtime Requests	—	Yes	Yes
Leave Requests	—	Yes	Yes
Holidays	—	—	Yes

8.2 Page Descriptions

Page	Route	Description
My Attendance	payroll/attendance/my-attendance/index.tsx	Today's punch status (IN/OUT buttons), current month calendar grid, recent punch history, pending correction requests. Shows: schedule for today, last punch time, daily summary (present/late/OT).
Attendance Sheets	payroll/attendance/sheets/index.tsx	Admin view: branch-scoped. Date picker, employee list with daily status (Present/Late/Absent/Holiday/On Leave). Click row to expand detail (time_in, time_out, late_mins, OT_mins, daily_wage). Manual override button per employee.
Corrections	payroll/attendance/corrections/index.tsx	Tabbed: Pending Approved Denied All. Each row: employee name, date, correction type, reason, status badge, [Approve] [Deny] buttons. Deny modal requires denial_reason. Badge counter on sidebar for pending count.
Schedules	payroll/attendance/schedules/index.tsx	List view of all schedule templates. Form: schedule_name, start/end time, work_days (checkboxes Mon–Sun), effective_from/to. Assign to employee via searchable dropdown.
Overtime Requests	payroll/attendance/overtime/index.tsx	Tabbed: Pending Approved Denied. Form: employee search, date, hours, reason. Auto-resolves shift_type based on schedule+holiday data.
Leave Requests	payroll/attendance/leaves/index.tsx	Tabbed: Pending Approved Denied. Form: date, leave_type dropdown, duration (full/half morning/half afternoon), is_paid checkbox, reason.
Holidays	payroll/attendance/holidays/index.tsx	Superadmin only. Calendar view + list. Add/edit holiday: name, date, type (regular/special), recurring_yearly toggle.

9. Integration Points

Integration	Detail
Audit Logging	Use existing <code>Payroll\Audit\Traits\Auditable</code> trait in all mutation controllers. Every create/update/approve/deny/void action generates an audit log record with before/after diffs.
Branch Scoping	Follow <code>EmployeePolicy</code> pattern. Superadmin = global. Admin = branch_id filter. Staff = own records only. All queries apply branch scope dynamically via policies, never inline role checks.
DB Transactions	All multi-table writes (log punch + process sheet, approve correction + create time_logs + re-process, generate period + lock sheets + create items) wrapped in <code>DB::transaction()</code> . Pessimistic locking (<code>lockForUpdate()</code>) on concurrent operations.
Inertia.js	All page responses use <code>Inertia::render()</code> with typed props matching TypeScript interfaces in <code>resources/js/types/</code> . Existing pattern: controllers return named props consumed by page components.
Pagination	Follow existing pattern: <code>paginate(50)->withQueryString()</code> on all list endpoints.
Date Handling	All date displays in TSX views use <code>toManilaTime()</code> from <code>@/utils/dateHelper</code> . All server-side computation uses <code>Asia/Manila</code> timezone via Carbon.
Payroll Sidebar	Add "Attendance" dropdown to <code>payroll-sidebar.tsx</code> . Update existing layout to accommodate new route group.
Existing Employee > Salary Bridge	<code>ProcessDailyAttendance</code> reads <code>employee.current_daily_rate</code> when computing <code>daily_wage</code> . The existing salary history tracking is not affected by attendance — it's read-only at computation time.
Future: Payroll Run Engine	<code>Payroll_period_items</code> table is designed to feed a subsequent payroll run engine that computes final <code>net_pay</code> including government deductions (SSS, PhilHealth, Pag-IBIG), cash advances, and 13th-month provisions.

10. Implementation Sequence

Build order organized by dependency. Each phase includes migrations, models, enums, controllers, requests, policies, services, routes, frontend pages, and Pest tests.

Phase	Scope	Key Deliverables
Phase 1: Config	company_configurations table + seeder + config editor page	All attendance/payroll parameters stored and admin-configurable via UI. Foundation for all later phases.
Phase 2: Schedules	employee_schedules table + CRUD + assignment UI	Admin can create shift templates and assign to employees with effective dates and work_days.
Phase 3: Time Logs	time_logs table + self-service punch portal + admin manual log + 5-min throttle	My Attendance page live. Staff can punch IN/OUT. Admin can manually create time_logs. Duplicate throttling active.
Phase 4: Core Engine	AttendanceService + attendance_sheets table + batch processing + admin sheets view	processDailyAttendance() fully implemented. Attendance sheets computed and viewable. Batch branch processing operational.
Phase 5: Corrections	attendance_correction_requests + submit/approve/deny + superior-only policy	Staff can submit correction requests. Admins review and approve. Approved corrections auto-create time_logs and re-process sheets.
Phase 6: OT + Leave	overtime_requests + leave_requests + approval workflows	OT requests with shift_type-based rate selection. Leave requests with full/half day duration. Both integrate into processDailyAttendance.
Phase 7: Holidays	holidays table + CRUD + integration into engine	Regular and special holidays with day-before rule. Holiday pay computed in processDailyAttendance.
Phase 8: Payroll Run	payroll_periods + payroll_period_items + period generation + void	Admin generates period, locks sheets, approves period. Superadmin can void. Period items feed future payroll computation engine.
Phase 9: Tests	Pest feature tests for all phases	Branch scoping, approval chain, grace period, 3-hour block, OT cross-reference, holiday day-before, leave blending, lockdown/void.

11. Edge Cases & Error Handling

11.1 Time Log Edge Cases

Scenario	Behavior
Two IN punches within 5 minutes	Earliest kept as canonical. Later marked as duplicate_of → canonical. ProcessDailyAttendance uses non-duplicate logs only.
Two OUT punches within 5 minutes	Same throttling as above. Earliest OUT kept.
Only IN, no OUT (day still in progress)	No attendance sheet generated. "In progress" status shown on portal. Not flagged as absent.
Only IN, no OUT (day closed)	Flagged as unexcused absence. Attendance_sheet created with 0 hours, notes = "Missing OUT punch."
Only OUT, no IN	Anomaly. Sheet created with warning. Cannot determine worked duration — 0 hours credited.
Punch exactly at midnight	Assigned to the previous day since no night shifts in the system.
Punch on a rest day	Self-service punch blocked unless pre-approved OT exists. Admin can manually log rest day work.
Punch outside schedule bounds (>18h after start)	Flagged as anomaly for admin review. Still logged but sheet marked with warning.

11.2 Computation Edge Cases

Scenario	Behavior
Late exactly at grace period boundary (e.g., 15 minutes late, grace = 15)	late_minutes = 0 (within grace).
Late at 16 minutes, grace = 15, forfeit = true	late_minutes = 16 (full forfeiture).
Late at 45 minutes, cap = 30	chargeable_late = 30. Deduction = $30 \times 10 = \text{₱}300$ max.
Holiday on Monday, employee absent Friday	Check Friday (last working day before). If absent without leave on Friday → unworked holiday pay = 0%.
Holiday on Monday, Friday was also a holiday	Walk further back to Thursday (last non-holiday working day).
Rest day work, employee works 2 hours	Paid for 2 hours at rest day OT rate (1.30). No 3-hour block penalty on rest days.
OT approved for 3 hours, employee stays 2 hours	Lower-of-two rule: approved_overtime_minutes = $\min(180, 120) = 120$ minutes.
OT approved for 1 hour, employee stays 2 hours	approved_overtime_minutes = $\min(60, 120) = 60$ minutes. Extra 60 minutes discarded.
OT approved but OT < threshold (e.g., 45 min, threshold = 60)	approved_overtime_minutes = 0. Must meet minimum consecutive threshold.
Schedule change mid-week	Employee_schedules.effective_from/effective_to determines which schedule applies for each date. ProcessDailyAttendance uses the most recently effective schedule.
Correction for already-locked sheet (paid payroll period)	Blocked. User must request superadmin to void the payroll period first.
Duplicate correction request (same	Blocked at application level. Only one pending correction per (employee_id, attendance_date, correction_type).

employee, date, type)

Self-approval attempt Policy blocks: submitter role must be strictly lower than reviewer role.

11.3 Concurrency & Locking

Scenario	Behavior
Two admins approve same correction simultaneously	<code>lockForUpdate()</code> on the correction request row ensures only one succeeds.
Payroll period generation running while sheets are being recomputed	Period generation locks sheets (<code>locked_at</code>) within a transaction. Any in-flight sheet recomputation fails if <code>locked_at</code> is already set.
Batch process for 200+ employees	<code>chunk(50)</code> processing with per-employee isolation. A failure on one employee does not roll back the entire batch.

12. Refinements from v1

#	v1 Issue	v2 Resolution
1	Late deduction was uncapped (infinite ₱10/min)	Cap at 30 minutes. Added late_deduction_max_minutes config. Max daily late penalty = ₱300.
2	No approval hierarchy on corrections	Superior-only chain: staff→admin, admin→superadmin. Self-approval blocked. Role comparison enforced in policy.
3	Manual attendance scope unclear	Staff: self-only. Admin: anyone in branch. Superadmin: anyone.
4	Multiple punch pairs unspecified	PH standard: single pair. First IN + last OUT only. No interval summing. Gaps = employee responsibility.
5	Holiday day-before rule naive (just check previous calendar day)	Walk backward to last scheduled working day. Skip rest days, Sundays, and other holidays. Holiday on Monday → check Friday.
6	OT rates hardcoded (1.25 only)	Moved to company_configurations. Four separate rates: regular (1.25), rest day (1.30), regular holiday (2.60), special holiday (1.30).
7	Only one OT field on attendance_sheets	Split into raw_overtime_minutes + approved_overtime_minutes. Full audit trail of actual stay vs awarded OT.
8	Rest days not accounted for	is_rest_day + is_rest_day_worked flags. Rest day work exempt from 3-hour penalty. Self-punch blocked on rest days.
9	Batch processing order dependency not addressed	Process dates in ascending order. Ensures holiday day-before sheets exist when processing a holiday date.
10	Correction table missing requester field	Added requested_by (FK→users.id). Separates "whose record" (employee_id) from "who submitted" (requested_by).
11	Leave blending not fully specified	Detailed blending matrix. Half-day morning + worked afternoon ≥ 4hr = full day. Half-day + insufficient worked = partial.
12	OT shift_type and rate snapshot missing	Added to overtime_requests. shift_type and ot_rate captured at approval time for audit trail.
13	Payroll lockdown ambiguity	Permanently locked once consumed. Superadmin voids entire period (unlocking all sheets) to allow corrections. No individual sheet unlocking.

14	Rest day partial work penalty ambiguous	Paid at rest day rate for actual minutes, no minimum. 3-hour block only applies to scheduled work days.
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End of Document

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